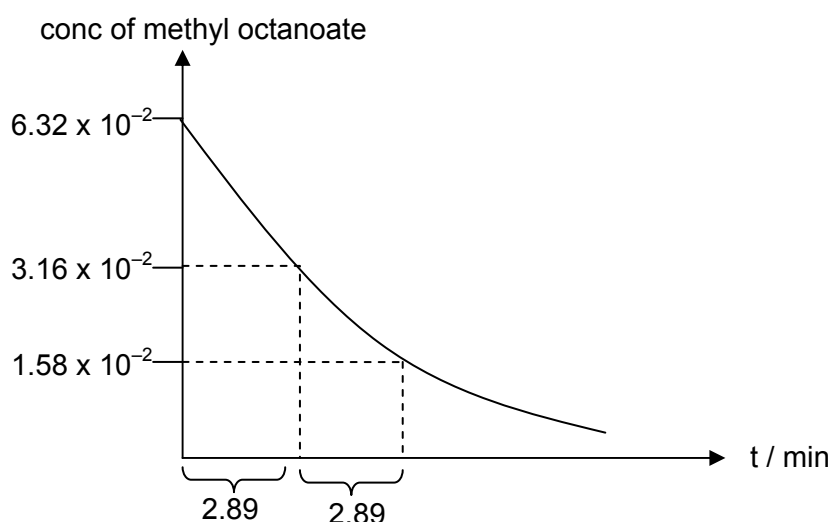


- 1 (a) (i) $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3(\text{CH}_2)_6\text{COOH} + \text{CH}_3\text{OH}$
- (ii) To prevent oxidation of methyl octanoate. / To prevent the oxidation of CH_3OH formed.
- (iii) Since rate constant has a unit of min^{-1} and water is used in large excess, order of reaction wrt $\text{CH}_3(\text{CH}_2)_6\text{COOH}$ is 1.

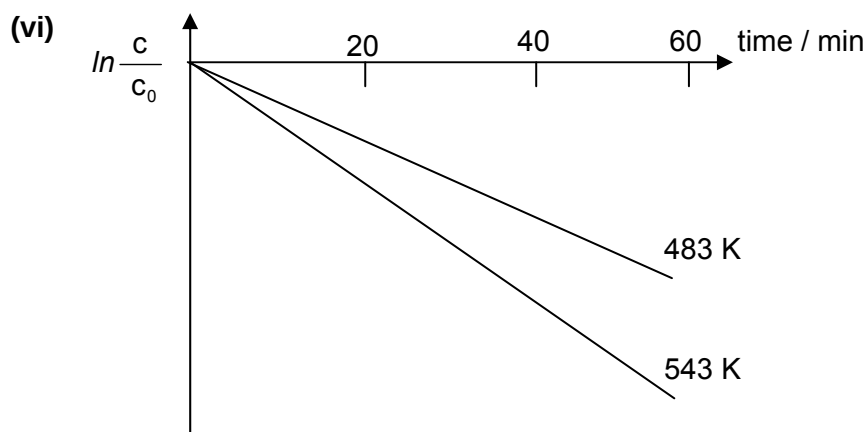
OR

Graph of $\ln \frac{c}{c_0}$ against t is a straight line. Hence, order of reaction is 1st order wrt $\text{CH}_3(\text{CH}_2)_6\text{COOH}$.

(iv) $t_{1/2} = \frac{\ln 2}{0.24} = 2.89 \text{ min}$



- (v) The rate of hydrolysis of $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$ is slower than that of $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$. Steric hindrance of the longer acyl chain slows down the nucleophilic attack by water.
- The activation energy of the hydrolysis of $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$ is higher than that of $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$.



As temperature increases, the average kinetic energy of the molecules increases. More molecules possess energy greater than or equal to the activation energy for the reaction.

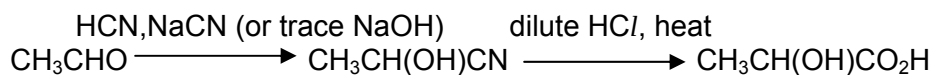
The frequency of effective collision increases. Rate of hydrolysis increases. Hence, $\ln \frac{c}{c_0}$ becomes more negative as temperature increases.

- (b) (i) Let the % of α -form in the equilibrium mixture be x .

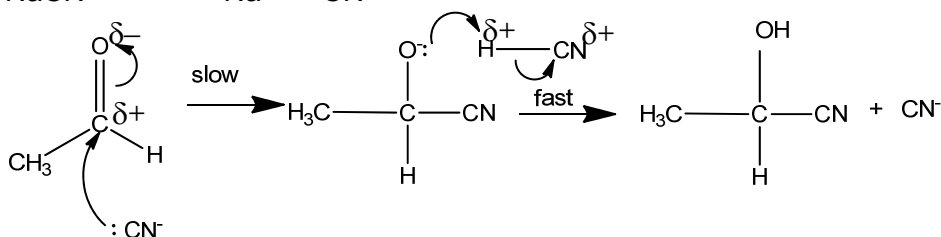
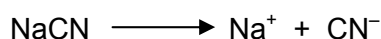
$$\frac{x(92.6) + (100 - x)(34)}{100} = 52.3$$

$$x = 31.2\%$$

- (ii)



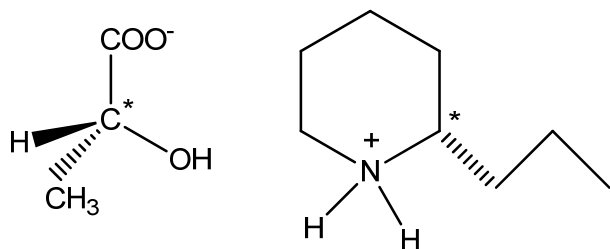
- (iii) Nucleophilic addition



- ✓ electron flow
- ✓ partial charge
- ✓ correct intermediate
- ✓ slow and fast steps

Electron deficient carbonyl carbon in ethanal is planar, hence it is equally susceptible to nucleophilic attack from either direction, resulting in a racemic mixture.

- (iv)



- 2 (a) (i)

$$K_p = \frac{p_{\text{SO}_3}^2}{p_{\text{SO}_2}^2 p_{\text{O}_2}} \quad \text{atm}^{-1} \quad \text{or} \quad \text{Pa}^{-1}$$

- (ii) To increase the SO_3 yield.

When the no. of moles of O_2 gas increases, the position of equilibrium will shift to the right by Le Chatelier's principle, and the resultant mixture will have more SO_3 and less SO_2 and O_2 .

- (iii) I. rate at which equilibrium is achieved.

When temperature is increased, the rate of the forward and backward reaction will increase. Hence the rate at which equilibrium is achieved will be faster.

II. equilibrium constant K_p

When temperature is increased, the rate of the backward reaction will increase to a greater extent than the rate of the forward reaction. Hence K_p will decrease.

- (b) (i) Guanidine is a strong base.
For the titration of H_2SO_4 (a strong acid) against guanidine, the pH of equivalence point is about 7.

(ii) $[\text{H}^+] = 10^{-0.70} = 0.20 \text{ mol dm}^{-3}$

$[\text{H}_2\text{SO}_4] = 0.10 \text{ mol dm}^{-3}$

- (iii) From the diagram, volume at end point = 34.0 cm^3 (accept 33.5 or 34.5)

Amount of H^+ used = $2(0.1 \times 25/1000) = 0.0050 \text{ mol}$

No. of moles of guanidine = 0.0050 mol

$[\text{guanidine}] = 0.0050 / (34/1000) = 0.147 \text{ mol dm}^{-3}$

(iv) molar mass of guanidine = $\frac{8.68}{0.147}$
 $= 59.0 \text{ g mol}^{-1}$

M_r of guanidine = **59.0**

- (c) (i) at MBC, $\text{pH} = \text{p}K_a$
 $= -\lg(6.3 \times 10^{-8})$
 $= 7.20$

- (ii) equivalence point is between 7.2 and 12.4

phenolphthalein is a suitable indicator as its working range is between pH 8 – 10, which coincides with the pH at equivalence point.

(iii) $K_b = 10^{-14} / 6.3 \times 10^{-8}$
 $= 1.59 \times 10^{-7} \text{ mol dm}^{-3}$

It is a stronger base because the K_b value of HPO_4^{2-} is greater than the K_a value of HPO_4^{2-} hence it has a greater tendency in accepting proton than donating proton.

(d) (i) $K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}] \text{ mol}^2 \text{ dm}^{-6}$

(ii) Concentration of barium sulfate solution = $\sqrt{1.08 \times 10^{-10}}$
 $= 1.04 \times 10^{-5} \text{ mol dm}^{-3}$

- (iii) BaSO_4 is less soluble in $\text{Ba}(\text{OH})_2$ due to the common ion effect (Ba^{2+}).
or

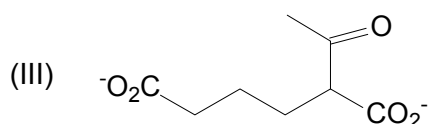
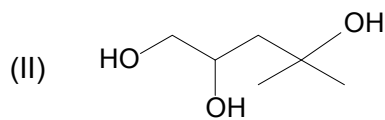
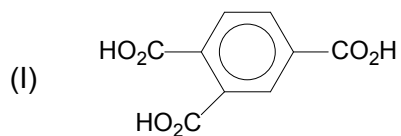


the position of equilibrium shifts to the left in the presence of barium hydroxide solution.

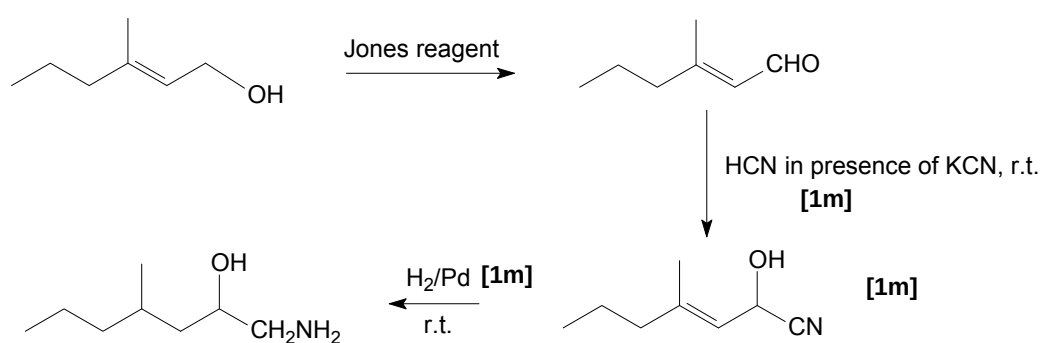
- 3 (a) (i) dark green solution: MnO_4^{2-} or K_2MnO_4
brown precipitate: MnO_2

(ii) colour change: purple to colourless (faint pink)
due to formation of colourless Mn^{2+}

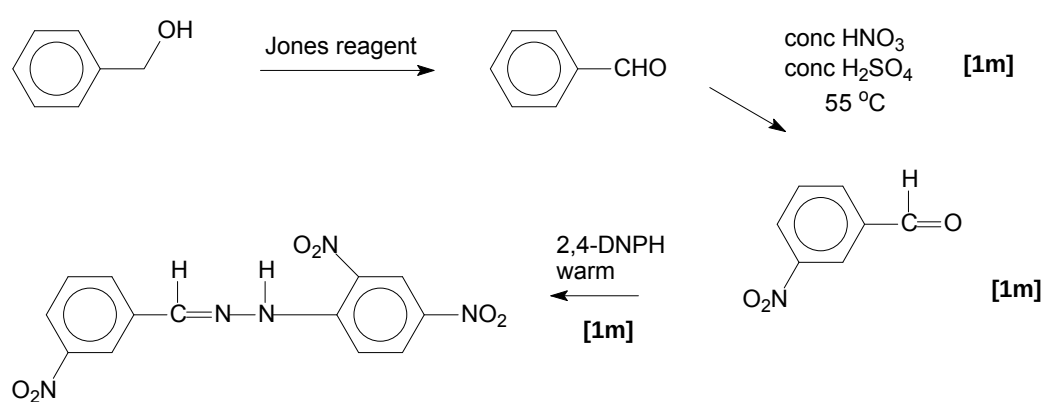
(iii)



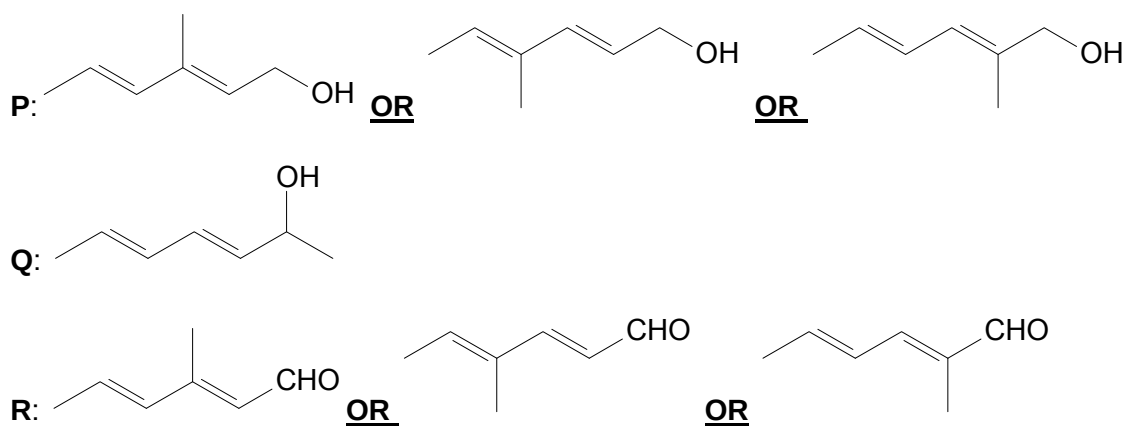
(b) (i)

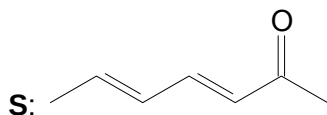


(ii)



(c)



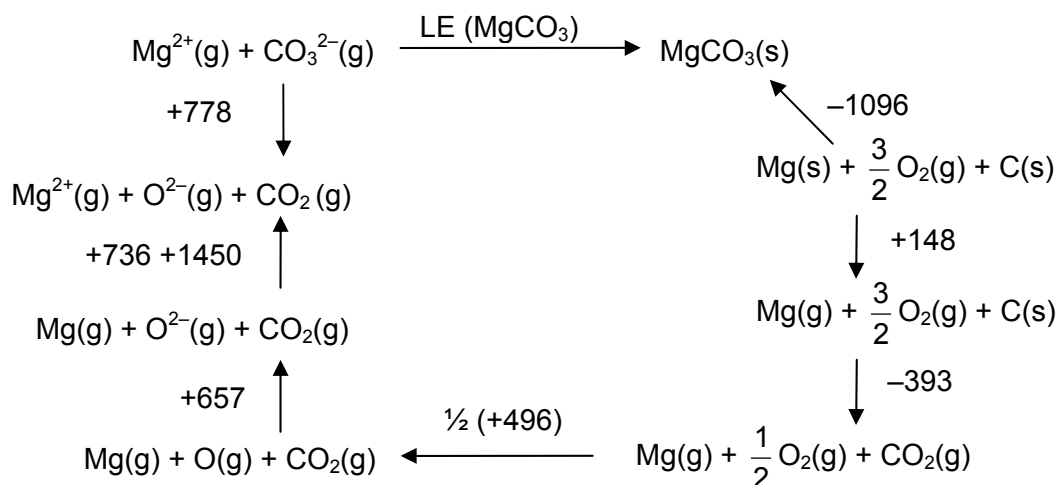


- (d) (i) chromic acid: +6
 dihydroxy(oxo) chromium: +4
 change in oxidation number = +4 – (+6) = –2

(ii) $\Delta H = [\text{BE}(\text{Cr}=\text{O}) + \text{BE}(\text{C}-\text{O}) + \text{BE}(\text{C}-\text{D})] - [\text{BE}(\text{C}=\text{O}) + \text{BE}(\text{O}-\text{D}) + \text{BE}(\text{O}-\text{H})]$
 $= (478 + 360 + 340) - (740 + 430 + 460)$
 $= -452 \text{ kJ mol}^{-1}$

- 4 (a) (i) It is the energy released when 1 mole of a solid ionic compound is formed from its constituent gaseous ions under standard conditions.

(ii)

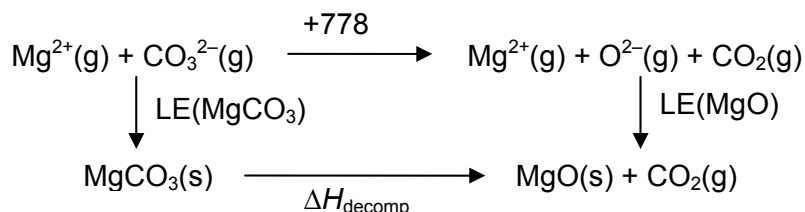


By Hess's Law,

$$\begin{aligned}
 \text{LE} &= +778 - (736 + 1450) - (+657) - \frac{1}{2}(+496) - (-393) - (+148) + (-1096) \\
 &= -3164 \text{ kJ mol}^{-1}
 \end{aligned}$$

- (iii) High charge density of Mg^{2+} distorts the electron cloud of CO_3^{2-} , thus the bonding in the lattice contains considerable covalent character, and makes the experimental value more exothermic than the theoretical value.

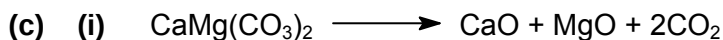
(iv)



$$\begin{aligned}
 \Delta H_{\text{decomp}} \text{ of } \text{MgCO}_3 &= -\text{LE}(\text{MgCO}_3) + 1189 + \text{LE}(\text{MgO}) \\
 &= -(-3164) + 778 + (-3800) \\
 &= +142 \text{ kJ mol}^{-1}
 \end{aligned}$$

- (b) (i) MgO , CO_2 , CaCO_3

- (ii) Since only 1 mol of gas was produced when 1 mol of $\text{CaMg}(\text{CO}_3)_2$ decomposed partially, only one of the carbonates in $\text{CaMg}(\text{CO}_3)_2$ had decomposed to form CO_2 . CaCO_3 had not decomposed yet as Ca^{2+} has a larger size, smaller charge density and weaker polarising power than Mg^{2+} . Electron cloud of CO_3^{2-} is distorted to a smaller extent and C–O covalent bond in CO_3^{2-} is weakened to a smaller extent by Ca^{2+} .

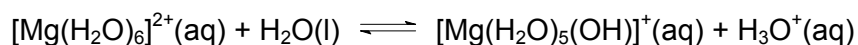
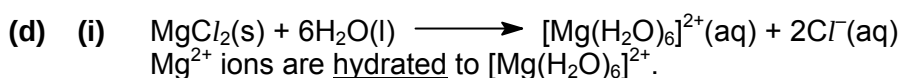


(ii) $n(\text{CaMgO}_2) = 700 \div [40.1 + 24.3 + 2(16.0)]$
 $= 7.261 \text{ mol}$

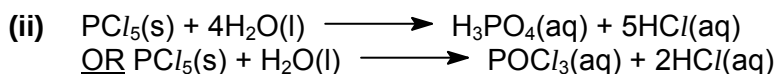
$$\Delta H_{\text{decomp}} \text{ of } \text{CaMg}(\text{CO}_3)_2 = 2142 \div 7.261 = +294.8 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{decomp}} \text{ of } \text{CaCO}_3 = +294.8 - (+142)$$

$$= +153 \text{ kJ mol}^{-1}$$

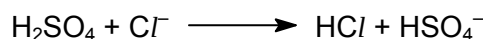


Mg^{2+} has high charge density and slightly polarises the O–H bond in water molecules, which is weakened and easily broken to form a slightly acidic solution.

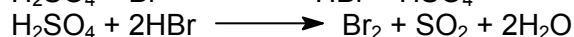
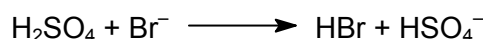


Not likely, since a strongly acidic solution is formed when PCl_5 reacts with water in the body and the excessive H^{+} formed may cause gastric problem.

- (iii) Br^{-} is a stronger reducing agent than Cl^{-} .

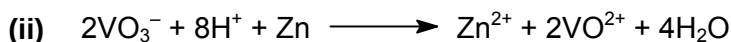


For MgCl_2 , white fumes of HCl are observed.

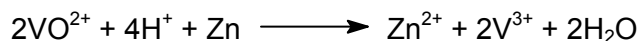


For MgBr_2 , white fumes of HBr and some orange/brown fumes of Br_2 , are observed.

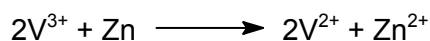
- 5 (a) (i) In the presence of water ligands, the partially filled 3d orbitals in vanadium ions are split into two levels (non-degenerate) with a small energy gap (correspond to visible light region)
 Energy is absorbed from the visible region when an electron promotes from a lower energy level d orbital to a vacant higher energy level d orbital, i.e. d–d transition.
 The colour observed is the complement of the colour absorbed.



$$E^{\circ}_{\text{cell}} = 1.00 - (-0.76) = +1.76 \text{ V} > 0$$



$$E^{\circ}_{\text{cell}} = 0.34 - (-0.76) = +1.10 \text{ V} > 0$$



$$E^{\circ}_{\text{cell}} = -0.26 - (-0.76) = +0.50 \text{ V} > 0$$

$$E^{\circ}_{\text{cell}} = -1.20 - (-0.76) = -0.44 \text{ V} < 0$$

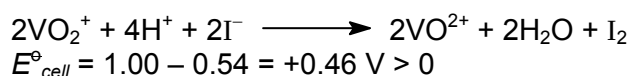
From the $E^\ominus_{\text{cell}}$ calculations, VO_3^- will be reduced by Zn to V^{2+} , and the final colour of the solution will be violet.

$$\begin{aligned} \text{(iii)} \quad n(\text{NH}_4\text{VO}_3) \text{ reacted} &= \frac{3.00}{(14 + 4 + 50.9 + 16 \times 3)} = 0.0257 \text{ mol} \\ n(\text{SO}_2) &= \frac{1}{2} (0.0257) \\ &= 0.01285 \text{ mol} \end{aligned}$$

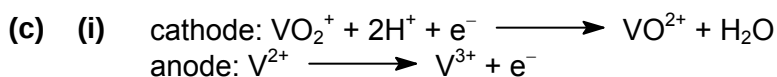
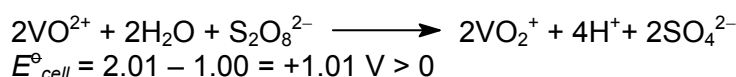
$$\begin{aligned} \text{Volume of SO}_2 \text{ needed} &= 0.01285 \times 24000 \\ &= \mathbf{308 \text{ cm}^3} \end{aligned}$$

(b) (i) A homogeneous catalyst is a catalyst that is in the same phase as the reactants.

(ii) Step 1: Formation of intermediate



Step 2: Regeneration of the catalyst



$$\begin{aligned} \text{(ii)} \quad E^\ominus_{\text{cell}} &= 1.00 - (-0.26) \\ &= \mathbf{+1.26 \text{ V}} \end{aligned}$$

(iii) This helps to maintains electrical neutrality (prevent build up of charges in the two half cells) so that electrical energy can continue to flow.

$$\text{(d)} \quad Q = It = 14 \times 20 \times 3600 = 1.008 \times 10^6 \text{ C}$$

$$n(\text{VO}^{2+}) = n(\text{e}^-) = 1.008 \times 10^6 / 96500 = 10.45 \text{ mol}$$

$$\begin{aligned} [\text{VO}^{2+}] \text{ initially} &= 10.45/5 \\ &= \mathbf{2.09 \text{ mol dm}^{-3}} \end{aligned}$$

(e) (i) Any $E^\ominus$ value that is more negative than that of $E^\ominus_{\text{V}^{3+}/\text{V}^{2+}} = -0.26\text{V}$ and its corresponding $E^\ominus_{\text{cell}}$ value correctly calculated.

(ii) Larger minimum external e.m.f needed to recharge the cell.